

Build Scalable Cloud Storage with Riak CS

Cloud storage is designed for large scale high technology environments. Data is stored in data centres run by third parties, which lease or sell storage space to clients. This article discusses the advantages of cloud storage, covering the services offered by various cloud storage providers, and concludes with the building of a scalable cloud storage system with Riak CS.



For some organisations, finding enough storage space to hold data acquired over a period of time is truly a challenge. Some companies invest in large storage systems that cost a fortune, while the rest bank on more traditional yet expensive methods such as magnetic disks and tape drives. But then, it's not only about the cost but also about a host of other factors, such as reliability, fault tolerance and on-demand scalability that make most traditional storage methods and systems obsolete.

The answer to all these issues? Cloud storage. Under this, an organisation's data is stored on an offsite distributed storage system that is run and maintained by a third-party provider. This data can be accessed remotely from any corner of the world at any time, as most cloud storage providers offer Web-based REST APIs that can be accessed over the Internet. And with an almost infinite storage capacity, organisations can now scale their storage needs

from megabytes to petabytes in just a matter of minutes; such is the power of cloud storage.

The advantages

The pay-as-you-go model: By using cloud storage, organisations have only to pay for the storage they consume. There are no upfront payments or long term investments required compared to traditional storage systems.

Scalability and availability: Traditional storage systems have hard limits on capacity and performance, which makes them difficult to scale. On the other hand, cloud storage providers offer dynamic scaling of storage resources that can grow and shrink based on the application's requirements. Cloud storage also offers geographically distributed storage capabilities that ensure the data is made available even in case a single storage server or an entire data centre in a particular region goes down.